

A function (函數) of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where n is a positive integer, is called a polynomial (多項式) in one variable (變元). The numbers $a_n, a_{n-1}, \dots, a_1, a_0$ are called the coefficients (係數) of the polynomial.

Further, if $a_n \neq 0$, then 'n' is the degree (次) of the polynomial, denoted by $\deg f$.

Two polynomials $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ and $g(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0$ are equal if and only if $m = n$ and all the corresponding coefficients are equal, and we write $f(x) \equiv g(x)$.

Operations on polynomials (多項式的運算)

Define 1. $f(x) \pm g(x) = (a_n \pm b_n) x^n + (a_{n-1} \pm b_{n-1}) x^{n-1} + \dots + (a_1 \pm b_1) x + (a_0 \pm b_0)$

2. $k(f(x)) = k a_n x^n + k a_{n-1} x^{n-1} + \dots + k a_1 x + k a_0$

3. $f(x) \cdot g(x) = a_n b_m x^{n+m} + (a_{n-1} b_m + a_n b_{m-1}) x^{n+m-1} + \dots + (a_1 b_0 + b_1 a_0) x + a_0 b_0$

4. $f^m(x) = (a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0)^m$, where m is a positive integer.

Example 1: Let $f(x) = x^4 + x^3 - 3x^2 + x + 2$. Prove that for any positive integer n , the polynomial $f^n(x)$ has at least one negative coefficient.

Division Algorithm (多項式的除法)

Suppose $\deg f(x) \geq \deg g(x)$, when $f(x)$ is divided by $g(x)$, we obtain the quotient $Q(x)$ and the remainder $R(x)$, where $\deg R(x) < \deg g(x)$. That is, $f(x) = g(x)Q(x) + R(x)$

Example 2: Divide $x^4 + 4x^3 - 18x^2 + 15x - 5$ by $x^2 - 3x + 2$.

Example 3: Divide $5x^4 + 3x^3 + 18x - 15$ by $x + 2$.

Example 4: Divide $16x^3 + 8x^2 - 2x + 1$ by $2x - 1$.

Example 5: Find the remainder when $x^{60} - 2x^{25} + 1$ is divided by $x^2 - 1$.

Remainder Theorem (餘數定理)

Theorem: The remainder when $f(x)$ is divided by $(x-b)$ is $f(b)$.

Proof: Suppose $f(x) = (x-b)Q(x) + R$, where $\deg(R) = 0$.

Putting $x = b$ gives $f(b) = R$.

Factor Theorem (因式定理)

Given the polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, $a_n \neq 0$. The binomial $(x - c)$ is a factor of $f(x)$ if and only if $f(c) = 0$.

Example 6: Factorize $f(x) = 6x^4 + 13x^3 + 3x^2 + 4x + 4$.

Example 7: Prove that $x(x+1)(2x+1)$ is a factor of $(x+1)^{2m} - x^{2m} - 2x - 1$.

Example 8: Given that a, b, c, d are positive integers, prove that $f(x) = x^{4a} + x^{4b+1} + x^{4c+2} + x^{4d+3}$ is divisible by $x^3 + x^2 + x + 1$.

Example 9: Prove that $(a+b)(b+c)(c+a) + abc = (a+b+c)(ab + bc + ca)$

Example 10: Prove that $(a+b+c)^3 - a^3 - b^3 - c^3 = 3(a+b)(b+c)(c+a)$

Factorization of Homogeneous Polynomials (齊次多項式的因式分解)

Example 11: Factorize $F(a, b, c) = a^3 + b^3 + c^3 - 3abc$

Example 12: Prove that $(a+b)(b+c)(c+a) + abc = (a+b+c)(ab + bc + ca)$

Example 13: Factorize $F(a, b, c) = \sum (b - c)(b + c)^3$.

Example 14: Factorize $F(a, b, c) = (a + b + c)^5 + (a - b - c)^5 + (b - c - a)^5 + (c - a - b)^5$.

Example 15: Factorize $F(a, b, c) = \sum_{abc} a^2(b^3 - c^3)$

Example 16: Factorize $F(a, b, c, x) = \sum_{abc} (a - b)^3(x - c)^2$

Example 17: (45th IMO 2014 No. 2) Find all polynomials f with real coefficients given that for all real numbers a, b, c such that $ab+bc+ca = 0$ we have the following relationship $f(a-b) + f(b-c) + f(c-a) = 2f(a+b+c)$.

Eisenstein Criterion (愛森斯坦判別法)

(Named after Gotthold Eisenstein, 1823-1852, a German Mathematician)

If $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, a_2, \dots, a_n$ are all integers, and there

exists a prime number p such that (i) $p|a_i$ for $i = 0, 1, 2, \dots, n-1$, (ii) $p \nmid a_n$, (iii) $p^2 \nmid a_0$,

then $f(x)$ is not reducible over the set of rational number. (不可約多項式).

Proof: Assume that there exist $g(x)$ and $h(x)$ such that $f(x) = g(x)h(x)$, and

$$g(x) = b_r x^r + b_{r-1} x^{r-1} + \dots + b_1 x + b_0, h(x) = c_s x^s + c_{s-1} x^{s-1} + \dots + c_1 x + c_0, r, s \in \mathbb{N}^+$$

We have $a_0 = b_0 c_0$. Since $p|a_0$ and $p^2 \nmid a_0$, hence w.l.o.g. we can assume that $p|b_0$ and $p \nmid c_0$.

On the other hand, since $p \nmid a_n$, and $a_n = b_r c_s$, hence $p \nmid b_r$,

Now consider $b_0, b_1, \dots, b_{r-1}, b_r$, starting from b_0 , we have the coefficients divisible by p , until we come to a certain b_k , where $0 < k < r$.

(i.e., $p|b_0, p|b_1, p|b_2, \dots, p|b_{k-1}, p \nmid b_k, p \nmid b_{k+1}, \dots, p \nmid b_r$)

Next consider $a_k = b_k c_0 + b_{k-1} c_1 + \dots + b_0 c_k$

Every term on RHS is a multiple of p except the first term, hence a_k is not divisible by p .

This is contradictory to condition (i).

Hence the assumption is wrong, and $f(x)$ is irreducible.

Example 18: The polynomial $f(x) = 2x^4 - 5x^3 + 10x + 15$ is irreducible.

Example 19: Prove that the polynomial $f(x) = 2x^4 + 9x^2 + 6$ is irreducible.

Example 20: The polynomial $f(x) = x^7 - 48x^2 + 24$ is irreducible.

Example 21: Prove that the polynomial $f(x) = x^2 - k$, where k is a prime number, is irreducible.

Note that this is a sufficient but not necessary condition.

For example, $f(x) = x^2 + 1$ is irreducible, but no suitable p can be found.

Example 22: Prove that the polynomial $f(x) = x^2 + x + 2$ is irreducible.

(Hint: Put $x = y+3$ to change the polynomial to $y^2 + 7y + 14$.)

Example 23: Prove that the polynomial $f(x) = x^4 + 1$ is irreducible.

Example 24 If p is a prime number, prove that the polynomial

$$f(x) = x^{p-1} + x^{p-2} + x^{p-3} + \dots + x + 1 \text{ is irreducible.}$$

Euclidean Algorithm(輾轉相除)

If $f(x)$ and $g(x)$ are polynomials of degrees n and m , both greater than 1. Suppose there exists a polynomial $h(x)$, unique apart from a constant factor, such that $h(x)$ is the Greatest Common Divisor of $f(x)$ and $g(x)$, then $h(x)$ can be found using the Method of Euclidean Algorithm, as follows:

Method: Given the polynomials $f(x)$ and $g(x)$, with all common constant factors removed beforehand.

Suppose $\deg f(x) \geq \deg g(x)$, then $f(x) = g(x) q_1(x) + r_1(x)$ -----(1)

$$g(x) = r_1(x) q_2(x) + r_2(x)$$

$$r_1(x) = r_2(x) q_3(x) + r_3(x)$$

$$r_2(x) = r_3(x) q_4(x) + r_4(x)$$

.....

$$r_{n-2}(x) = r_{n-1}(x) q_n(x) + r_n(x)$$

$$r_{n-1}(x) = r_n(x) q_{n+1}(x)$$

Finally, $r_n(x)$ is the G.C.D. of $f(x)$ and $g(x)$.

Proof: Consider (1). Let $d(x)$ be the Greatest Common Divisor (G.C.D.) of $f(x)$ and $g(x)$ and $d'(x)$ be the G.C.D. of $g(x)$ and $r_1(x)$.

Since $d'(x)$ be the G.C.D. of $g(x)$ and $r_1(x)$, then it is a factor of $f(x)$, and hence $d(x)$. Similarly by rewriting in the form $r_1(x) = g(x) q_1(x) - f(x)$, we see that since $d(x)$ is a factor of both $f(x)$ and $g(x)$, this means that $d(x)$ is a factor of $r_1(x)$, and hence a common factor of $r_1(x)$ and $g(x)$ and hence a factor of $d'(x)$.

From above we see that $d(x)$ and $d'(x)$ are factors of each other and they differ by at most a constant.

As common constant factors have been removed beforehand, therefore the G.C.D. of $f(x)$ and $g(x)$ is the same as the G.C.D. of $g(x)$ and $r_1(x)$.

Repeating the above procedure, we can say that the G.C.D. of $f(x)$ and $g(x)$, $g(x)$ and $r_1(x)$, $r_1(x)$ and $r_2(x)$, $r_2(x)$ and $r_3(x)$, ..., $r_{n-1}(x)$ and $r_n(x)$ are all the same.

Now $r_n(x)$ is a factor of $r_{n-1}(x)$.

Hence $r_n(x)$ is the G.C.D. of $r_{n-1}(x)$ and $r_n(x)$, and hence $f(x)$ and $g(x)$.

Example 25: Find the G.C.D. of $f(x) = 2x^4 - x^3 - 5x^2 + 13x - 12$ and $g(x) = 6x^4 - 5x^3 + 17x - 12$.

Example 26: Find the G.C.D. of $f(x) = x^4 - 5x^3 + 6x^2 + 4x - 8$ and $g(x) = x^3 - x^2 - 4x + 4$.

Example 27: Find the G.C.D. of $f(x) = 3x^4 + 15x^3 + 5x^2 + 10x + 2$ and $g(x) = 2x^4 + 9x^3 + 14x + 3$.

Bézout's Theorem for Polynomials (裴蜀定理)

(Named after Etienne Bezout, 1730-1783, a French Mathematician.)

If $f(x)$ and $g(x)$ are two polynomials with the G.C.D. $h(x)$, then there exist two other polynomials $F(x)$ and $G(x)$ such that $f(x)F(x) + g(x)G(x) = h(x)$.

By suitable choice we can always have $\deg F(x) < \deg g(x)$ and $\deg G(x) < \deg f(x)$.

For if $\deg F(x) > \deg g(x)$, then we can choose $\lambda(x)$ such that $F(x) - \lambda(x)g(x)$ is of degree less than $g(x)$, and obtain $f(x)[F(x) - \lambda(x)g(x)] + g(x)[G(x) + \lambda(x)f(x)] = 1$.

Here we have $F'(x) = [F(x) - \lambda(x)g(x)]$ and $\deg F'(x) < \deg g(x)$, and

$$G'(x) = [G(x) + \lambda(x)f(x)] \text{ and } \deg G'(x) < \deg f(x).$$

In particular, if $f(x)$ and $g(x)$ are co-prime, then there exist polynomials $F(x)$ and $G(x)$ such that $f(x)F(x) + g(x)G(x) = 1$

Theorem: For any co-prime polynomials $f(x)$ and $g(x)$, if $F(x)$ and $G(x)$ are so chosen that $f(x)F(x) + g(x)G(x) = 1$, and $\deg F(x) < \deg g(x)$, $\deg G(x) < \deg f(x)$, then $F(x)$ and $G(x)$ are unique.

Proof: Suppose $F(x)$, $G(x)$ and $F_1(x)$, $G_1(x)$ satisfy the given conditions, that is $f(x)F(x) + g(x)G(x) = 1$ and $f(x)F_1(x) + g(x)G_1(x) = 1$, and $\deg F(x) < \deg g(x)$ and $\deg G(x) < \deg f(x)$, $\deg F_1(x) < \deg g(x)$ and $\deg G_1(x) < \deg f(x)$. then $[F(x) - F_1(x)]f(x) = [G_1(x) - G(x)]g(x)$. Now $f(x)$ and $g(x)$ are co-prime, hence $f(x)$ is a factor of $[G_1(x) - G(x)]$. However, $\deg [G_1(x) - G(x)] < \deg f(x)$. This leads to contradictions. The only possible case is that $G_1(x) = G(x)$, and similarly $F_1(x) = F(x)$. That is, the existence of $F(x)$ and $G(x)$ are unique.

Example 28: Find the G.C.D. of $f(x) = x^4 + 3x^3 + x^2 - 2$ and $g(x) = x^4 - x^3 + x^2 + 2$, hence express the G.C.D. in the form $f(x)F(x) + g(x)G(x)$.

Example 29: Show that $f(x) = x^3 + x + 1$ and $g(x) = x^2 - x + 1$ are co-prime, and hence express 1 in the form $f(x)F(x) + g(x)G(x)$ for some polynomials $F(x)$ and $G(x)$.

Example 30: Find a polynomial $p(x)$ such that (x^2+1) is a factor of $p(x)$, and $x^3 + x^2 + 1$ is a factor of $p(x) + 1$.

Root of a Polynomial (多項式的根)

Given a polynomial $f(x)$. If $f(\alpha) = 0$, then α is called a zero or a root of the polynomial.

Fundamental Theorem of Algebra (代數基本定理)

Every non-constant single variable polynomial has at least one root, which may be real or complex.

Theorem: A polynomial of degree n ($n \geq 1$) has exactly n roots, though some of the roots may be repeated roots, and the roots may be real or complex.

Proof: Let the polynomial be $f(x)$.

Part 1: Using the Fundamental Theorem of algebra, it has at least one root, say r_1 .

Then $f(x) = (x - r_1)g(x)$, where $g(x)$ is a polynomial of degree $n-1$.

Suppose $n-1 \geq 1$, then again it has at least one root, say r_2 .

So, $f(x) = (x - r_1)(x - r_2)h(x)$

The process may be repeated until we obtain $f(x) = a_n(x - r_1)(x - r_2)\dots(x - r_n)$, where a_n is the coefficient of x_n in $f(x)$.

Hence $f(x)$ has n roots

Note: If a root occurs twice, it is called a double root (二重根). If it occurs three times, it is called a triple root (三重根). If it occurs m times, it is called a root of multiplicity m (m 重根).

Part 2: Suppose a polynomial $f(x)$ of degree n has m roots, where $m > n$.

Now, take any n roots $r_1, r_2, \dots, r_n$, we have

$f(x) = a_n(x - r_1)(x - r_2)\dots(x - r_n)$.

Further, $f(x)$ is zero for at least one other root, say r_{n+1} .

Hence $f(r_{n+1}) = a_n(r_{n+1} - r_1)(r_{n+1} - r_2)\dots(r_{n+1} - r_n)$

We must have $a_n = 0$, and hence $f(x)$ is not a polynomial of degree n .

This is contradictory. Hence the assumption is wrong, and $f(x)$ has at most n roots.

Corollary: If $f(x)$ and $g(x)$ are polynomials of degree n , and $f(x_i) = g(x_i)$ with $x_1, x_2, \dots, x_m$ distinct numbers and $m > n$, then $f(x) \equiv g(x)$.

Proof: $f(x) - g(x)$ is a polynomial of degree less than or equal to n , but with m distinct roots, where $m > n$. Using the above Theorem, $f(x) - g(x) \equiv 0$, hence $f(x) \equiv g(x)$.

Example 31 Given that a, b and c are distinct real numbers, prove that

$$(a) \quad \frac{(x-b)(x-c)}{(a-b)(a-c)} + \frac{(x-c)(x-a)}{(b-c)(b-a)} + \frac{(x-a)(x-b)}{(c-a)(c-b)} = 1$$

$$(b) \quad \frac{a^2(x-b)(x-c)}{(a-b)(a-c)} + \frac{b^2(x-c)(x-a)}{(b-c)(b-a)} + \frac{c^2(x-a)(x-b)}{(c-a)(c-b)} = x^2.$$

Example 32: Given that $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ are $2n$ distinct real numbers, solve the following system of equations:

$$\begin{aligned} \frac{x_1}{a_1-b_1} + \frac{x_2}{a_1-b_2} + \dots + \frac{x_n}{a_1-b_n} &= 1, \\ \frac{x_1}{a_2-b_1} + \frac{x_2}{a_2-b_2} + \dots + \frac{x_n}{a_2-b_n} &= 1, \\ &\dots\dots\dots \\ \frac{x_1}{a_n-b_1} + \frac{x_2}{a_n-b_2} + \dots + \frac{x_n}{a_n-b_n} &= 1, \end{aligned}$$

Example 33: Prove that $(x+1)(x+2)(x+3)(x+4) + 1 = (x^2 + 5x + 5)^2$.

Example 34: Prove that $f(x) = (x-1)(x-2)(x-3)\dots(x-10) - 1$ is irreducible over the set of integers

Example 35: Let $f(x)$ be a polynomial of degree n and with integral coefficients. Prove that if there are at least $2n+1$ distinct integers m such that $|f(m)|$ is a prime number, then $f(x)$ is irreducible over the set of integers.

Method of Comparing Coefficients (係數比較法)

Example 36: Find the values of a and b so that $f(x) = x^4 + ax^3 + bx^2 - 8x + 4$ is the square of a polynomial.

Example 37: Find the sum to n terms of the series $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2$.

Example 38: Find the sum to n terms of the series $1^4 + 2^4 + 3^4 + \dots + n^4$

Example 39: Find the sum to n terms of the series $1^3 + 3^3 + 5^3 + \dots$

Example 40: (17th IMO 1975 Number 6.)

Find all polynomials P , in two variables, with the following properties:

- (i) For a positive integer n and all real t, x, y , $P(tx, ty) = t^n P(x, y)$.
(that is, P is a homogeneous of degree n .)
- (ii) For all real a, b, c , $P(b+c, a) + P(c+a, b) + P(a+b, c) = 0$.
- (iii) $P(1, 0) = 1$.